

$$X_{n+1} = 2 X_n$$

$$X_0 = 5$$

$$X_n = 5 \cdot 2^n$$

Eigenvector Eigenvalue

$$Av = \lambda v$$

numpy `la.eig(A)`

rank of A $\text{rank} = \# \text{ cols} - \dim \text{Null}$

`matrix_rank(A)`

`la.svd(A)`

0 singular values

$\text{Null}(A) =$ correspond rows of V_h

$$V_0 \quad V_{n+1} = A V_n$$

$$\text{if } Au = 2u$$

$$\lambda = 2$$

eigenvector u

$$\text{if } V_0 = u$$

$$\text{then } V_1 = Au = 2u$$

$$V_2 = A(2u) = 2Au \\ = 2^2 u$$

$$V_n = 2^n u$$

$$\text{if } \lambda = \frac{1}{2} \text{ then } V_n = \frac{1}{2^n} u$$

$$\text{as } n \rightarrow \infty, V_n \rightarrow 0$$

$$2. \quad \underset{n \times n}{A} \underset{n \times n}{B} = n \times n$$

$$\sum_{k=1}^n a_{ik} b_{kj}$$

$$(AB)C$$

$$\sum_{m=1}^n \left(\sum_{k=1}^n a_{ik} b_{km} \right) c_{mj}$$

$$= \sum_{m=1}^n \sum_{k=1}^n a_{ik} b_{km} c_{mj}$$

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\det A = ad - bc$$

$$\operatorname{tr} A = a + d$$

A^{-1} exists when
iff $\det A \neq 0$

$$\frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$XX^t = Y$ is Symmetric iff $Y = Y^t$

$6 \times 7 \times 7 \times 6$
 Y is 6×6

Simplify Y^t

$$= (XX^t)^t$$

two identities

$$(AB)^t = B^t A^t$$

$$(A^t)^t = A$$

$$= (X^t)^t X^t$$

$$= X \cdot X^t$$

$$= Y \text{ by def}$$

Since $Y^t = Y$, Y is Symmetric

Show Y has no nontrivial null vectors

Assume $Yv = 0$

$$X \cdot X^t v = 0$$

$$\langle X^t v, X^t v \rangle = 0$$

$$\langle X \cdot X^t v, v \rangle = 0$$

$$\Rightarrow X^t v = 0$$

X^t Nonnull
 $\Rightarrow v = 0$

Use $\langle Au, w \rangle$

$$= \langle u, A^t w \rangle$$

$$V \in \text{Null}(A)$$

iff

$$A V = 0$$

$$\begin{aligned} \text{Length of } V &= \sqrt{V \cdot V} \\ &= \sqrt{V \cdot V} \end{aligned}$$

$$V \cdot V \geq 0 \quad \text{only } 0 \text{ if } V=0$$

$$\begin{aligned} V &= (-1, 2) & V \cdot V \\ & &= (-1)^2 + 2^2 \end{aligned}$$